TRAFFIC SIGN DETECTION USING DEEP LEARNING

JUAN FELIPE ACUÑA
GUILLERMO HERNANDEZ
EMANUEL ALBERTO WILCHEZ

TABLE OF CONTENTS

01. Problem Definition
02. Models & Parameters
03. ResNet-50
04. EfficientNet-B4
06. Swin-Transformer (Swin-T)
07. Results Analysis
08. Conclusions
09. References

PROBLEM DEFINITION

Colombia reports 6,000–7,000 road deaths per year. A key cause: drivers ignoring or not recognizing traffic signs.
Our solution: An automatic traffic sign classifier that can be integrated into dashcams or mobile apps to alert drivers in real time.
Why GTSRB? Colombian signs follow the same international standards as German ones — same shapes, colors, and symbols. Data augmentation simulates tropical weather conditions.

PARAMETERS

All of the parameters described below are derived from two combined sources: the specific characteristics of the balanced GTSRB dataset (43 classes, 250 images per class, images resized to 224×224) and the recommendations in NVIDIA's official guide on convolutional layer performance

The batch size is set to 64 for all three models. NVIDIA states that the performance of the Tensor Cores improves when the batch size is a multiple of 64, as this allows for full parallelization of the GPU's Streaming Multiprocessors.

With 7,525 training images (70% of the balanced dataset), a batch size of 64 generates approximately 117 steps per epoch, which is sufficient for the gradient to be representative of the 43 classes at each update.

NVIDIA explicitly recommends this format because the convolutions implemented for Tensor Cores require it natively.

MODELS

-ResNet-50 -EfficientNet-B4 -Swin-Transformer (Swin-T)

RESNET-50

IMAGE PROCESSING

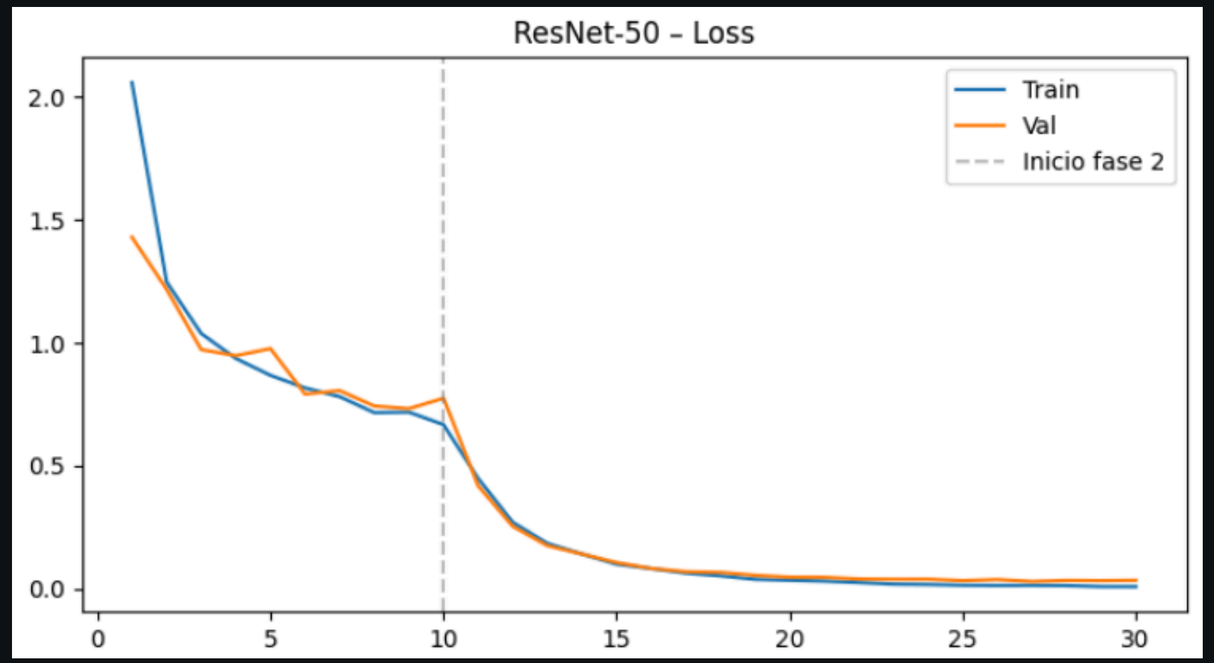
Phase 1 — 10 epochs:
Frozen backbone
- Head only
- Adam LR 0.001

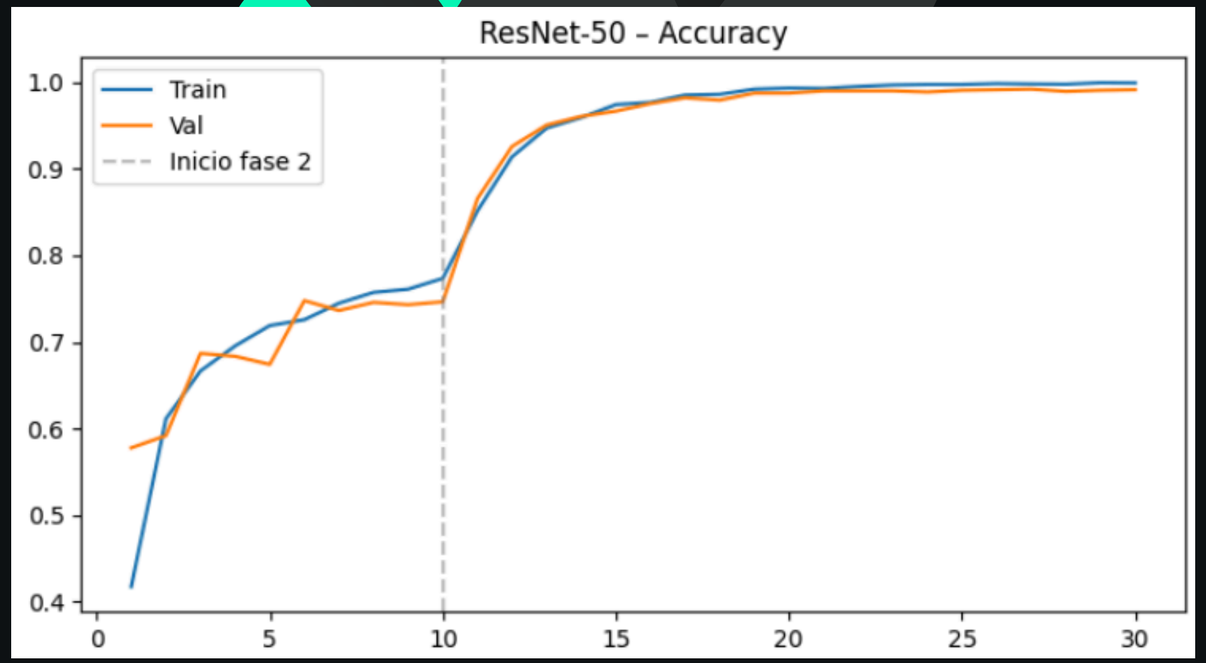
Phase 2 — up to 20 epochs: Full fine-tuning with layer-wise learning rates:

Layer group	LR
Layers 1–2 (edges, textures)	0.000001
Layer 3 (shapes)	0.00001
Layer 4 (complex patterns)	0.00005
Classifier head	0.001

Early stopping: no improvement in 6 epochs stop and restore best model.

RESULTS RESNET-50





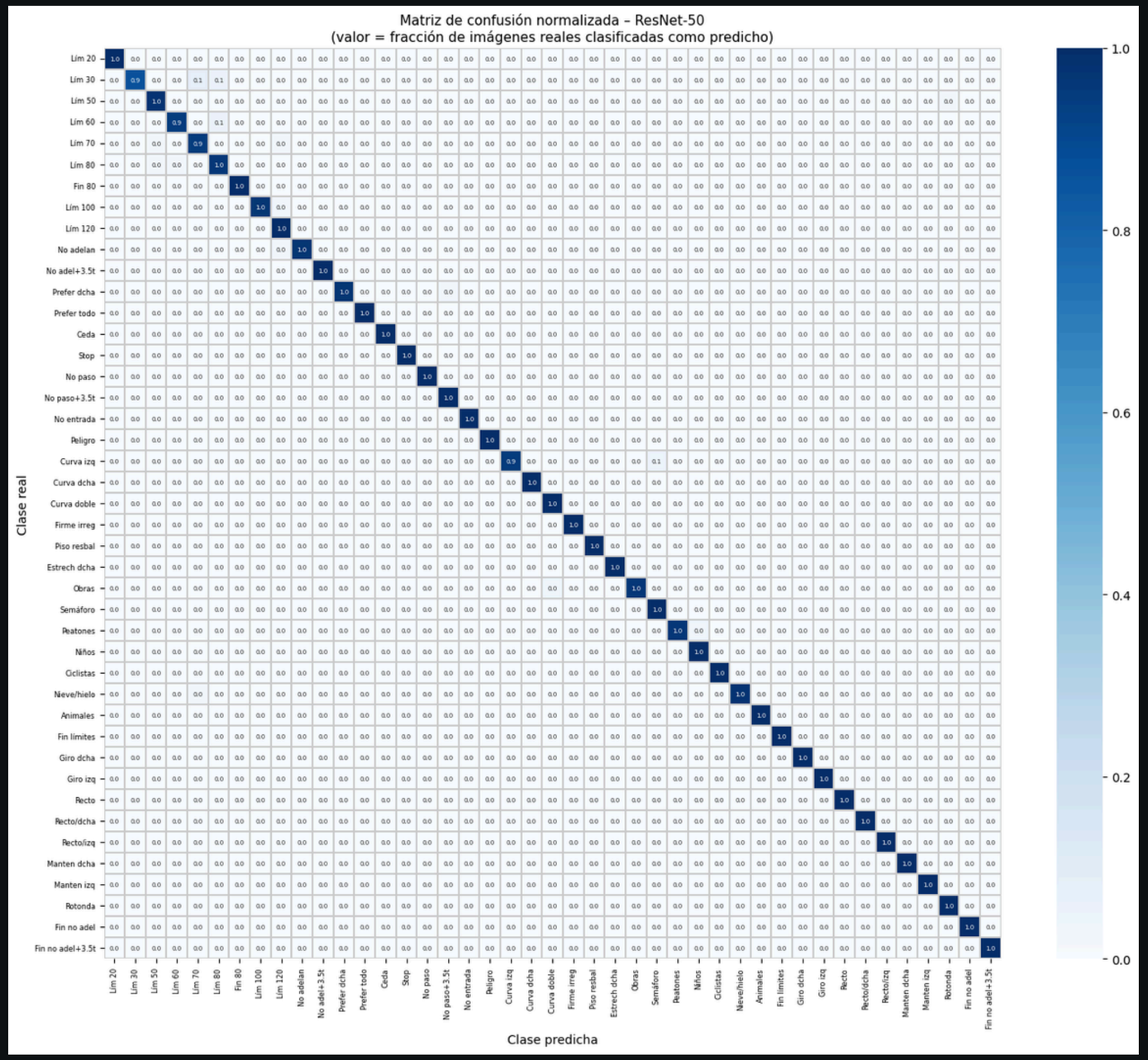
LOSS FUNCTION

Loss starts high at ~2.0 (train) and ~1.4 (val) in epoch 1. During Phase 1 it drops steadily, reaching ~0.7 by epoch 10. Once Phase 2 begins, the decline accelerates sharply. By epoch 20 the loss is near 0 and stays stable until epoch 30. Train and validation curves run together the whole time — no overfitting detected.

ACCURACY

In epoch 1, training accuracy starts at 40% and validation at 58%, showing that ImageNet pretraining already provides useful knowledge. During Phase 1 both curves rise gradually to ~75% by epoch 10. Once Phase 2 begins, accuracy jumps rapidly from 75% to over 95% in just 3-4 epochs. From epoch 20 onward it climbs smoothly to near 100%, where both curves completely overlap — confirming no overfitting.

RESULTS RESNET-50



CONFUSION MATRIX

The diagonal line indicates that the vast majority of the 43 classes were classified correctly. Errors are rare and do not show any patterns of widespread confusion between groups.

The model achieved an overall accuracy of 98.73%. Most of the 43 classes achieved high accuracy; the only classes with slight drops were those that were visually very similar to one another, all of which corresponded to speed limit signs.
Total training time: 33 minutes

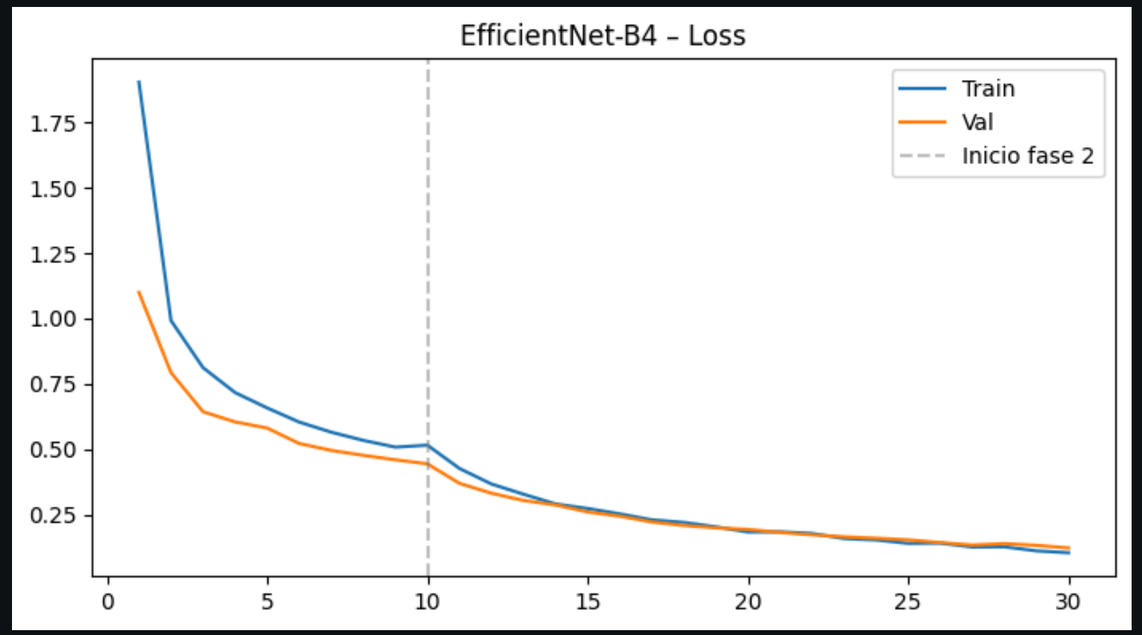
EFFICIENTNET-B4

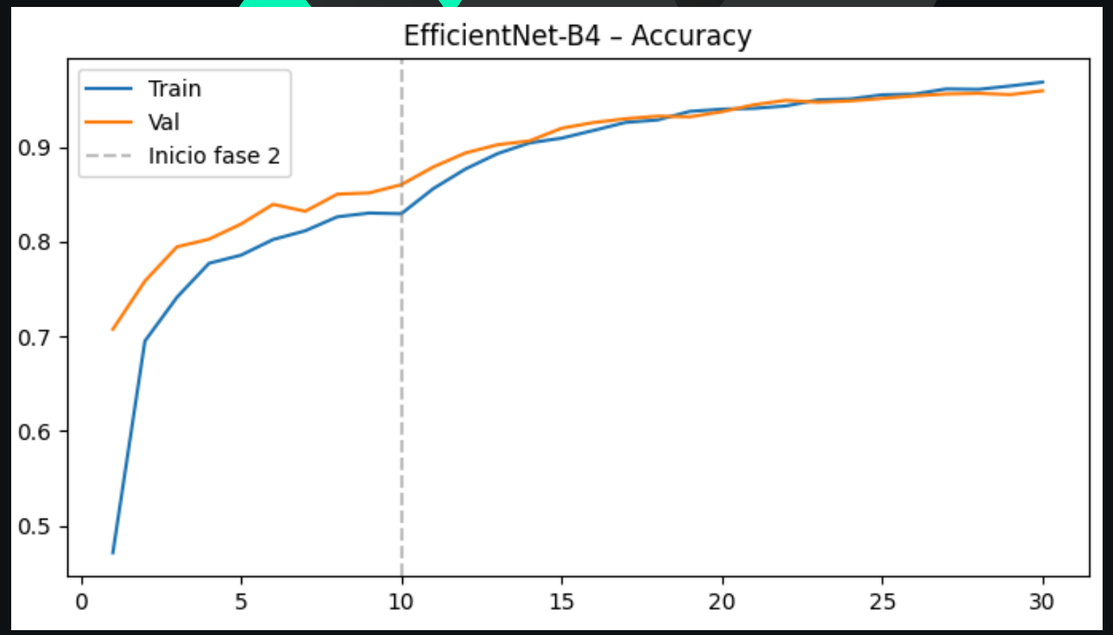
IMAGE PROCESSING

Phase 1:
10 epochs with a frozen backbone:
Train only the head. Adam with a learning rate of 5e-4 (lower than in ResNet due to EfficientNet's sensitivity).
A label smoothing of 0.1 is used to prevent overfitting.

Phase 2 – up to 20 epochs with progressive fine-tuning
Layer group LR
Layers 1–2 (edges, textures) 0.000001
Layer 3 (shapes) 0.00001
Layer 4 (complex patterns) 0.00005
Classifier head

RESULTS EFFICIENTNET-B4





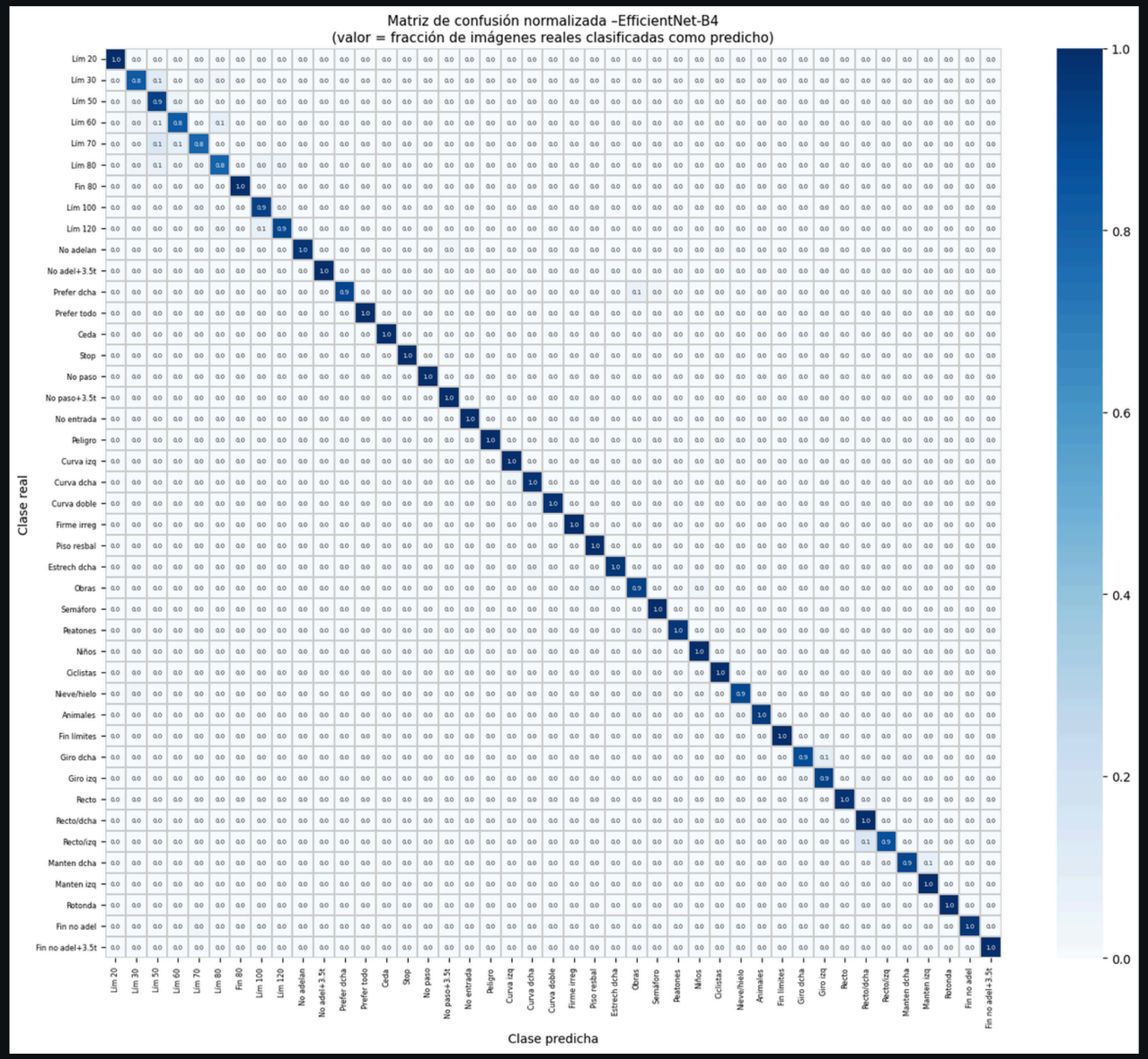
LOSS FUNCTION

Loss starts at 1.85 (train) and 1.1 (val). Unlike ResNet-50, validation drops faster during Phase 1, showing that EfficientNet's pretrained knowledge transfers more easily. By epoch 10 both curves sit around 0.5. In Phase 2 they decline smoothly to ~0.15 by epoch 30, more gradual than ResNet — reflecting that conservative learning rates produce slower but more stable learning.

ACCURACY

Training starts at 47% but validation already opens at 71%, confirming that EfficientNet extracts richer ImageNet representations. During Phase 1 validation climbs to 84% while training gradually catches up. In Phase 2 both curves rise steadily in parallel, stabilizing near 95%. Unlike ResNet-50 which reached 99%, EfficientNet levels off lower and the curves never fully overlap — suggesting it could still improve with more epochs or additional tuning.

RESULTS EFFICIENTNET-B4



CONFUSION MATRIX

The diagonal is dominant in most classes, with an accuracy of 95.31%. Errors are concentrated in three areas:
Speed limits
Right-of-way signs
Directional signs
EfficientNet handles classes with unique shapes and colors well, but it has a harder time than ResNet distinguishing numbers within circular signs with only 250 images per class.

The model achieved 95.31% accuracy. This is a solid result, but lower than the 98.73% achieved by ResNet-50, because EfficientNet-B4 has a more modern architecture. There are more problematic classes than in ResNet, and they are particularly concentrated around speed limits.
Total training time: 42 minutes

SWIN-TRANSFORMER

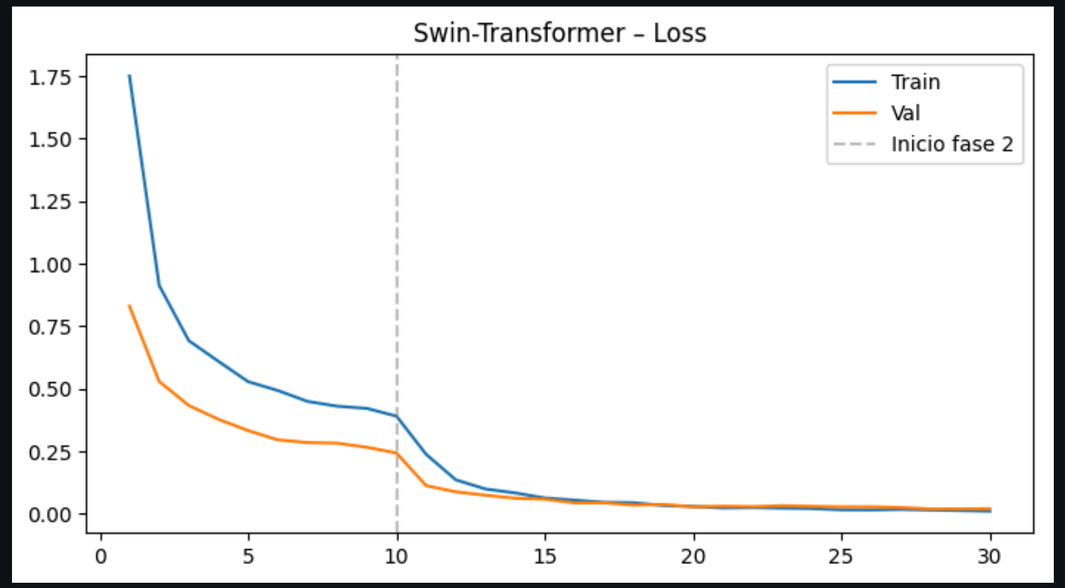
IMAGE PROCESSING

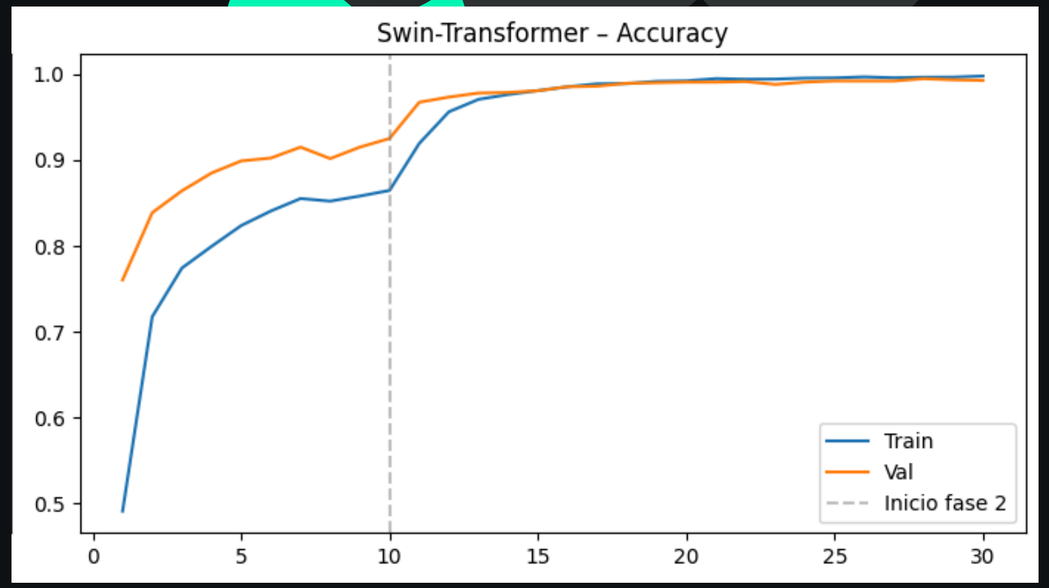
Phase 1 – 10 epochs with a frozen backbone: Train only the head. Learning rate 0.0001 (the lowest of the three, due to the sensitivity of Transformers). Adam optimizer.

Phase 2 – up to 20 epochs with fine-tuning
Key differences: AdamW (the standard for Transformers) is used, along with linear warmup during the first 2 epochs (the learning rate gradually increases from 0 to the target value), because all four stages are unfrozen simultaneously and a high value from the start could destabilize the attention matrices.

layers	Learning rate
Stages 1 and 2 (early local care)	0,000001
Stage 3 (intermediate setting)	0,000005
Stage 4 (global setting)	0,00002
Classification header	0,0001

RESULTS SWIN-TRANSFORMER





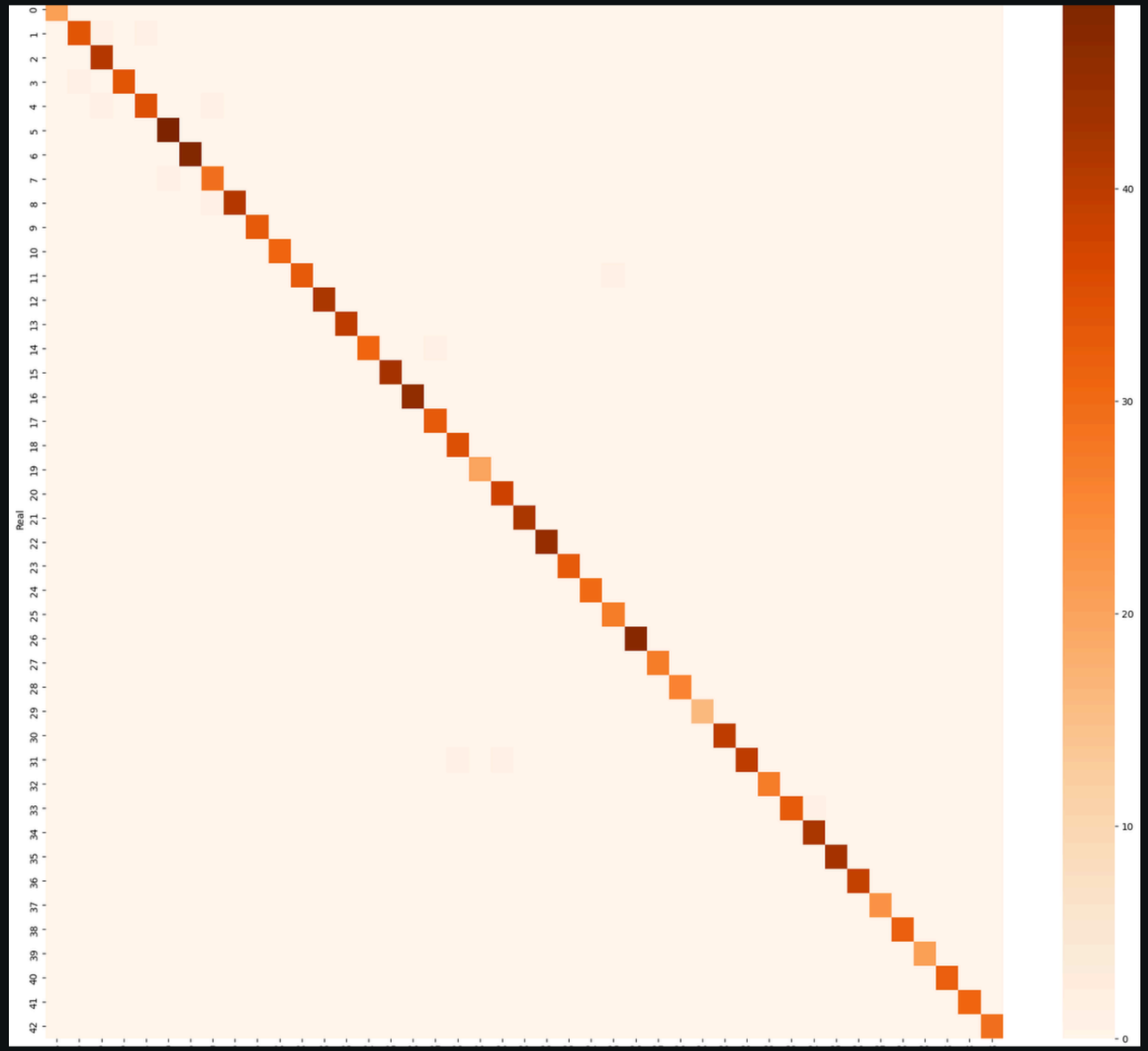
LOSS FUNCTION

Starts high (~1.75 train, ~0.85 val) and drops quickly in early epochs, showing effective learning. During phase 1 both curves decrease steadily and stay close. In phase 2, loss drops further and stabilizes near ~0.02, indicating strong convergence.

ACCURACY

Training begins around 49% while validation is higher (~76%), showing good initial transfer. Both increase steadily to ~86–92% by epoch 10. In phase 2, they rise and overlap near 99–100%, indicating high performance without clear overfitting.

RESULTS SWIN-TRANSFORMER



CONFUSION MATRIX

The matrix is heavily dominated by the diagonal, indicating an overall accuracy close to ~99%, which reflects an almost perfect classification across the 43 classes. Unlike models such as ResNet-50 (98.73%) or EfficientNet-B4 (95.31%), this result suggests that Swin-Transformer achieves a higher level of precision and consistency. Errors are extremely sparse and appear only in a few isolated off-diagonal cells, without forming clear clusters of confusion between specific groups of classes.

DATA ANALYSIS

ResNet-50

Achieved a high accuracy of ~98.73%, with both training and validation curves showing a strong upward trend and eventual convergence. The loss function decreases rapidly, especially after the second training phase, indicating effective learning. The confusion matrix shows a dominant diagonal.

EfficientNet-B4

Reached an accuracy of ~95.31%, with smoother and more gradual improvements in both loss and accuracy curves. The validation accuracy starts higher than training, indicating good initial generalization, but both stabilize below the other models. The confusion matrix still shows a clear diagonal, though with more noticeable off-diagonal values

Swin Transformer

Obtained the highest accuracy (~99–100%), with training and validation curves that quickly converge and remain almost identical throughout the process. The loss drops consistently to very low values, showing stable optimization. Its confusion matrix is nearly perfectly diagonal, with almost no off-diagonal values, indicating minimal classification errors and the most consistent performance among the three models.

References

[1] German Traffic Sign Benchmark, "GTSRB Dataset," acceso: 9 de marzo de 2026. Disponible en: https://benchmark.ini.rub.de/gtsrb_dataset.html
[2] J. Stallkamp, M. Schlipsing, J. Salmen y C. Igel, "The German Traffic Sign Recognition Benchmark: A multi-class classification competition," The 2011 International Joint Conference on Neural Networks, 2011, doi: 10.1109/IJCNN.2011.6033395.
[3] J. Stallkamp, M. Schlipsing, J. Salmen y C. Igel, "Man vs. computer: Benchmarking machine learning algorithms for traffic sign recognition," Neural Networks, 2012, doi: 10.1016/j.neunet.2012.02.016
[4] A. Jaiswal, Deepali y N. Sachdeva, "Empirical Analysis of Traffic Sign Recognition using ResNet Architectures," 2023 3rd International Conference on Advance Computing and Innovative Technologies in Engineering, 2023, doi: 10.1109/ICACITE57410.2023.10183247.
[5] A. Baruah, R. Kumar y M. Gupta, "Traffic Sign Recognition Using Deep Learning Neural Network and Spatial Transformer," 2023 International Conference on Advances in Computing, Communication and Applied Informatics, 2023, doi: 10.1109/ACCAI58221.2023.10199560.
[6] A. Farzipour, O. N. Manzari y S. B. Shokouhi, "Traffic Sign Recognition Using Local Vision Transformer," 2023 13th International Conference on Computer and Knowledge Engineering, 2023, doi: 10.1109/ICCKE60553.2023.10326288. [7] Q. Luo y W. Zheng, "Pre-Locator Incorporating Swin-Transformer Refined Classifier for Traffic Sign Recognition," Intelligent Automation & Soft Computing, 2023, doi: 10.32604/IASC.2023.040195. [8] X. Lu, "Zero-shot contrastive vision-language pre-training for traffic sign recognition in adverse weather conditions," International Conference on Computer Vision, Robotics, and Automation Engineering, 2024, doi: 10.1117/12.3042011. [9] X. H. Lim, W. Kin Lai y M. K. Lim, "Composite-Scaled Efficientnet for Enhanced Malaysian Traffic Sign Recognition," 2025 International Conference on Machine Learning and Cybernetics, 2025, doi: 10.1109/ICMLC66258.2025.11280101